

Ho Ko

AI/ML SOFTWARE ENGINEER

CONTACT

EMAIL

woodyhoko@gmail.com

LOCATION

San Francisco Bay Area, CA

WEBSITE

hoko.xyz

LINKEDIN

linkedin.com/in/hoko

GITHUB

github.com/woodyhoko

SKILLS

Languages

Python

C/C++

Swift

Java

Kotlin

JavaScript

x86 Assembly

MATLAB

ML Lifecycle

Model Training

Distillation

Quantization & Compression

Inference Engines

On-Device Deployment

Scalable Serving

On-Device & Web

WebGPU

Three.js

Transformers.js

MediaPipe

WebRTC

Local LLMs (Gemma, LFM)

Domains

NLP

Computer Vision

Healthcare

Astronomy

Weather Forecasting

Economics

EDUCATION

- M.S. Computer Science

USC — AI & ML

2021–2023 · GPA 4.0/4.0
- M.S. Electrical & Computer Eng.

Waterloo — ML & AI

2020–2021 · GPA 3.9/4.0
- B.S. Computer Science

NCU, Taiwan

2016–2020 · GPA 3.8/4.0

AI/ML Software Engineer with cross-domain experience spanning astronomy, weather forecasting, economics, healthcare, NLP, and computer vision. Works across the full **model lifecycle** — training and distilling deep models, architecting highly efficient inference engines, and building scalable, high-throughput serving — and brings that intelligence **on-device** through WebGPU, Three.js, and local LLMs in privacy-preserving systems that have reached **1B+ users**.

EXPERIENCE

- Machine Learning Engineer · Google

2026–Present

 - Build **knowledge-grounded retrieval & ranking models** on Google Lens that power AI Mode in Google Search and AI Overviews at Search scale.
 - Distill Gemini** into highly compressed models that hold near-equal quality while meeting the Lens stack's latency, scalability, and multimodal requirements.

- Software Engineer · Google

2023–2026

 - Core contributor to the **LiteRT LM inference engine**, running LLMs privately and offline across Windows, macOS, Linux, and Android — including NPU acceleration.
 - Enabled Gemma multimodality (vision + audio + text) and Gemini Nano on-device; helped ship **Chrome Built-in AI** and **Google AI Core**, reaching **1B+ users**.
 - Shipped the **Google AI Edge Gallery** app on Android and iOS.

- Research Collaborator · Google DeepMind

2024–2025

 - Designed **Sparse RAG (ICLR 2025)**, a sparse decoding paradigm that encodes retrieved docs in parallel and attends only to the most relevant caches via learned control tokens.
 - Cut long-range attention latency while **improving generation quality across 4 retrieval benchmarks**.

- Research Collaborator · Google Health & Fitbit

2024–2025

 - Built and evaluated ML models **detecting hypertension risk from raw wrist PPG** passively collected on a consumer smartwatch (medRxiv), enabling non-invasive screening at population scale.

- Software Engineer · MyCareLinq

2023

 - Built **NLP pipelines** and a **home-health-care knowledge graph** from unstructured health documentation.
 - Integrated LLMs powering caregiving assistants that surface contextual recommendations and care-plan summaries.

- Software Engineer Intern · Google

2022

 - Built a TFLite tensor extractor + metadata-writer with classification rules at **100% coverage of common GenAI task libraries** (LiteRT Task Library).
 - Integrated a **C++ code generator** compiling optimized inference pipelines for arbitrary TFLite files.

- Research Assistant · Caltech (SURF)

2018

 - Analyzed transient imaging from the Zwicky Transient Facility; **co-authored a Type IIb supernova study in ApJL (878:L5)** and simulated double-peak light-curve models.

SELECTED PUBLICATIONS

- Accelerating Inference of Retrieval-Augmented Generation via Sparse Context Selection** — [ICLR 2025](#), Google DeepMind.
- Opportunistic Hypertension Detection from Wearable PPG** — [medRxiv 2025](#), Google Health & Fitbit.
- A Type IIb Supernova (ZTF18aalrxas)** — [ApJL 878:L5](#), 2019.
- Real-world WebAssembly Performance for ML** — [ACM IMC 2023](#).